

Occupational Health and Safety – Short Notes

Based on the uploaded 50-page scanned safety document.

1. Accident, Hazard and Risk

- **Accident:** an unexpected and unplanned event that may cause injury, illness, property damage or environmental damage.
- **Hazard:** a source, energy, machine, material, activity or condition that can cause harm.
- **Risk assessment:** identify hazards, consider who may be harmed, assess likelihood and severity, and select suitable controls.

2. Hazard Identification & Control

- Inspect the workplace, equipment, materials and work methods regularly.
- Review previous accidents, incidents and near misses and talk with workers.
- Control hazards by removing them where possible, using safer methods, engineering controls/guards, administrative controls, training and supervision.
- PPE is an additional protective measure and must be selected for the hazard.

3. Machine Safety

- Keep hands, clothing and body parts away from moving or rotating parts.
- Keep guards and safety devices in place and correctly adjusted.
- Check equipment before use; do not use damaged equipment.
- Stop/isolate equipment before cleaning, maintenance or adjustment.
- Grinding: use suitable eye/face protection, check the grinding wheel and maintain the correct tool-rest clearance.

4. Manual Handling & Ergonomics

- Use correct lifting techniques; keep loads close to the body and avoid twisting.
- Use mechanical aids for heavy or awkward loads and ask for assistance when needed.
- Reduce repetitive work and awkward postures through suitable workstation arrangement, task variation and job rotation.

5. Personal Protective Equipment (PPE)

Main PPE categories in the source: **head, eye/face, hearing, hand, foot, respiratory, body and fall protection.**

- PPE must match the hazard, fit correctly, be inspected before use and be maintained/replaced when damaged.
- Safety helmets protect against falling objects/impact. Safety footwear may include protective toe caps, steel midsoles and slip-resistant soles.
- Hearing protection includes earplugs and earmuffs. Eye protection is required for hazards such as dust, particles, chemicals and sparks.
- Gloves should be selected according to the task and hazard; correct size improves comfort, control and protection.

Occupational Health and Safety – Short Notes (Page 2)

6. Respiratory Protection & Protective Clothing

- Respiratory protection can include filtering facepieces, half/full-face respirators and supplied-air or self-contained breathing equipment.
- Select respiratory protection according to the contaminant and exposure conditions.
- Protective clothing may be needed for chemical, biological, heat, splash and other workplace hazards.

7. Fall Protection

- Fall-protection equipment includes safety belts and safety harnesses.
- Use suitable fall-arrest/restraint equipment where there is a risk of falling from height.
- Inspect equipment before use and connect it to a suitable anchorage system.

8. Confined Space Safety

Confined spaces may have limited access, poor ventilation, oxygen deficiency, toxic gases, flammable gases, heat, engulfment and physical hazards.

Before entry: **identify hazards** → **isolate hazards** → **test the atmosphere** → **provide ventilation/controls** → **use PPE and communication** → **provide rescue arrangements** → **authorize entry**.

Only authorized personnel should enter. Atmospheric conditions should be monitored as required.

9. Important Gas & Fire Information

- **Fire triangle:** air/oxygen + flammable gas, vapour or dust + ignition source. Removing one element can prevent combustion.
- **Methane:** LFL $\approx$ 5% by volume; UFL $\approx$ 15% by volume; the source states that before confined-space entry, flammable gas should be below 10% of the LFL.
- **Hydrogen sulfide (H₂S):** permissible exposure limit 10 ppm (8-hour average); short-term exposure limit 15 ppm (15-minute average); IDLH 100 ppm.

10. Gas Sampling

Relative gas density shown in the source: Methane 0.55; Ammonia 0.59; Carbon monoxide 0.96; Nitrogen 0.97; Air 1.0; Hydrogen sulfide 1.2; Carbon dioxide 1.5; Gasoline 3–4; Jet Fuel JP-8 4.7.

Gases lighter than air tend to collect at higher levels; gases heavier than air tend to collect at lower levels. Sampling should therefore consider different heights.

11. Atmospheric Testing Instruments

- Oxygen deficiency → **oxygen analyzer**
- Flammable gases → **explosimeter**
- Toxic gases → **gas/vapour detector**; levels should be below applicable occupational exposure limits.

12. Confined-Space Isolation & Entry

Isolation methods shown in the source include **blanks and blinds, lockout and tagout, and double block and bleed**.

Entry should be controlled by the responsible supervisor/attendant, with the required permit, testing, ventilation, PPE, communication and rescue arrangements.

Where hazardous areas require it, use intrinsically safe (Ex) equipment and control ignition sources.

Quick Exam Reminder: Identify the hazard → assess the risk → control the hazard → use PPE → monitor the workplace → review controls.